Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/27/2011
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hollingsworth, Andrew (New York University)		
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	Colloid science is entering a new era. Over the past 15 years, our NASA-sponsored research has mainly dealt with monodisperse suspensions of colloidal particles interacting via well-known forces. Using spherical particles and observations with light scattering and microscopy, we have gained a great deal of fundamental knowledge about different phases of matter and the dynamics and thermodynamics of their formation. In particular, our experimental results in microgravity have lead to a basic understanding of why crystals and glasses form and their properties. During the past decade, we have made great strides in synthesizing new classes of particles with different shapes and specific, reversible or irreversible, variable range interactions. We have also found new ways to manipulate the particles with flow, electric and magnetic fields and light. We are therefore positioned at the threshold of a new technology, assembling equilibrium and non-equilibrium macroscopic structures with function and activity from well designed		

Task Description:	particles on the nano to micron scale. Of course, there are still fundamental scientific questions which we can and will address including a host of new ordered phases, frozen configurations, frustration and glasses and the process of self-organization itself. In particular, we plan to use the microscopy and light scattering instruments, in collaboration with our European colleagues, to study particles that we prepare through emulsion and dispersion polymerization. Physical lithographic techniques will also be employed, and the particles will be modified chemically for controllable interactions. We plan to use different phoretic techniques– electro-, dielectro-, and thermo-phoresis– to control the particles density and orientation. These will also serve as the driving forces to establish the rheological properties of these new systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterization of crystal formation in the microgravity environment of the ISS (International Space Station) can lead to a greater understanding of how gravity affects many kinds of colloidal materials, including monodisperse ellipsoids and cubes, colloidal clusters of silica or polymer microspheres, DNA-functionalized colloidal spheres, and 'lock-and-key' colloids. By performing these experiments in reduced gravity, we intend to accomplish the desired characterization without gravitationally-induced inhomogeneities that affect both the dynamics and equilibrium state on earth. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices to be used in optical communication systems. With the ability to make particles of different shapes, i.e., non spherical, we also have the possibility of having directionally dependent particle interactions. For example, we could take tetrahedral clusters of particles and attach DNA to them. The complementary single-stranded DNA 'sticky ends' can associate/dissociate via thermal activation. This arrangement could lead to tetrahedral bonding as found in diamond or in amorphous glass structures. Another approach utilizes depletion interactions. Since we can lithographically prepare particles of any shape we design in two dimensions and many shapes in three dimensions, we can fabricate lock-and-key colloids which only bind to their complementary shape. In this case, the binding is also directional since the congruent surfaces must match. We can also make such lock-and-key particles through emulsion chemistry. Our goal is to produce some simple processes with such ‘designer particles’ and interactions, to lay the foundations for self-assembly and perhaps self-replication of this new class of materials.
Task Progress:	Highlights: • Images of Sample 9, BCAT-5 were downloaded from ISS February 8 through March 23, 2011 (approximately 1,000 h observation time). With the present photos and image processing we were able to see evidence of colloidal crystals forming after about 4 – 8 days. We believe that photos taken at other angles will confirm that we are clearly seeing the effects of surface curvature (due to the seed particles) on colloidal crystal formation. We uploaded image processed photos that were taken by NASA astronaut Cady Coleman with the camera perpendicular to the sample module. This data will help optimize image quality during the next round of experiments with Sample 9 and greatly enhance the science return. • PACE2 samples: Wrote and submitted sample filling protocols to Zin Technologies. Sample filling was conducted at Zin in April 2010. • Chemical synthesis: Scaled up and successfully prepared a one liter quantity of PHS-g-PMMA comb graft stabilizer solution (40% nonvolatile) for PMMA colloidal particle synthesis. Published a journal article on this subject. Prepared and characterized PMMA particles for Canadian Space Agency experiment BCAT-C1 (with B. Frisken, Simon Fraser Univ.). Prepared and characterized 1.3 micron diameter (4% polydispersity) PMMA, crosslinked, RITC dyed particles for confocal microscopy studies. Prepared core-dyed PMMA particles for confocal microscopy studies and published a journal article on this subject. • Binary Colloidal Alloy Test (BCAT), Preliminary Advanced Colloids Experiments (PACE) The BCAT-6 sample module (including NYU samples 6 – 8), as well as the NYU PACE-2 sample disk (see April 2010 Progress Report on NASA Award Number NNX08AK04G) were launched February 24, 2011 on space shuttle Discovery’s STS-133 mission. Currently, BCAT-6 is a reserve payload and will be run in the future when time allows. The BCAT-6 investigation is structured from a rich history of space flight experiments that explore the fundamental physical science and application of colloids in a microgravity environment. Colloids are a type of heterogeneous mixture in which very small particles of one substance are distributed evenly throughout another substance. Paints, milk, fog, butter, smoke, ink, paint are colloids. This effort will address fundamental questions in colloidal engineering that impact product shelf life and determine how concentrated systems of particles of select sizes and shapes cause order to naturally arise out of disorder when gravity is removed. Our model system is comprised of synthetic polymer particles of uniform size and shape, dispersed in oil. In this experiment (BCAT-6 samples 6 and 7), half of the container surface is covered by large spherical seeds, while the other portion is not. Two volume fractions were prepared, one at the minimum theoretical volume fraction causing complete crystallization and one in the glassy regime. The ‘crystallization catalysts’ will allow us to do several experiments at once and see the effects of (1) placing seed particles on the cell walls, and (2) determining how the particle volume fraction can affect the rate of crystal nucleation. If this strategy is successful, we should be able to optimize experimental parameters when we study seeded growth later with the Advanced Colloids Experiment (ACE). While a significant effect is predicted theoretically, this experiment has been complicated because of particle sedimentation on earth. Thus, microgravity allows us to have both index matching (for clarity) and density matching (to avoid sedimentation in 1 g). White light from a flashlight and the camera flash will be used to detect the colorful Bragg reflections produced by colloidal crystals formed following the sample mixing. BCAT-6 sample 8 represents the first micro-gravity flight of self-annealing, DNA-coated polystyrene colloids. Thermally induced hybridization of the constituent linker strands will allow the particles to bind together, thus enabling them to explore self-assembly, reproduction, and mutation. The sample was designed such that small changes in cabin temperature will cause the particles to interact reversibly. Aggregated three-dimensional particle structures will melt

	(unbind) as the sample temperature increases; the individual particles will subsequently recombine as the ambient temperature decreases. We will look for changes in sample turbidity and Bragg reflections produced by any crystallites that may form during the thermal cycling. This type of suspension offers exponential growth—needed for next stage of micro- and nano-technology. In addition to providing an expanded toolkit to colloidal engineering, exponential growth provides information on competition and evolution, which can then be used to enhance colloidal self-assembly models. In preparation for the PACE experiments, we synthesized silica ellipsoids and cubes, which are two types of aspherical colloids. We have found that depending on the particular shape, ellipsoidal particles can pack more densely than spheres in both the crystalline and random configurations. Regarding the colloidal cubes, we have done experiments that show that crystallization can be induced with depletion agents (dissolved polymers); however the micron-size silica cubes are not density matched with the suspending medium. We will eliminate the dissolved polymer and study the crystallization and phase behavior of a concentrated sample in micro-gravity. Both types of colloid contain added salt to screen surface charges.
Bibliography Type:	Description: (Last Updated: 09/29/2025)
Articles in Peer-reviewed Journals	Elsesser MT, Hollingsworth AD. "Revisiting the synthesis of a well-known comb-graft copolymer stabilizer and its application to the dispersion polymerization of poly(methyl methacrylate) in organic media." Langmuir. 2010 Dec 7;26(23):17989-96. http://dx.doi.org/10.1021/la1034917 , Dec-2010
Articles in Peer-reviewed Journals	Elsesser MT, Hollingsworth AD, Edmond KV, Pine DJ. "Large core-shell poly(methyl methacrylate) colloidal clusters: synthesis, characterization, and tracking." Langmuir. 2011 Feb 1;27(3):917-27. http://dx.doi.org/10.1021/la1034905 , Feb-2011
Articles in Peer-reviewed Journals	Irvine WT, Vitelli V, Chaikin PM. "Pleats in crystals on curved surfaces." Nature. 2010 Dec 16;468(7326):947-51. http://dx.doi.org/10.1038/nature09620 , Dec-2010
Articles in Peer-reviewed Journals	Jaoshvili A, Esakia A, Porrati M, Chaikin PM. "Experiments on the random packing of tetrahedral dice." Physical Review Letters. 2010 May 7;104(18):185501. http://dx.doi.org/10.1103/PhysRevLett.104.185501 , May-2010
Papers from Meeting Proceedings	Chaikin PM. (invited speaker) "M&Ms Packing and Research with Undergrads." Session V5: Physics for Everyone. American Physical Society March Meeting, Dallas, Texas, March 21-25, 2011. American Physical Society March Meeting, Dallas, Texas, March 21-25, 2011. , Mar-2011
Papers from Meeting Proceedings	Chaikin PM. "Topological defects on Colloidal Crystals on Curved Surfaces." Corporate seminar, Rhodia Research Labs, Lyon, France, December 2010. Corporate seminar, Rhodia Research Labs, Lyon, France, December 2010. , Dec-2010
Papers from Meeting Proceedings	Chaikin PM. "Topological defects on Colloidal Crystals on Curved Surfaces." Physics of colloidal suspensions and granular media. SFB TR6 Summer School, Cargese Corsica, September 27 - October 09, 2010. Physics of colloidal suspensions and granular media. SFB TR6 Summer School, Cargese Corsica, September 27 - October 09, 2010. , Oct-2010
Papers from Meeting Proceedings	Chaikin PM. "Five lectures: Introduction to Colloids, Soft Matter Physics and Experimental Geometry Landau Theory of Phase Transitions Generalized Elasticity, Topological Defects." Scottish Doctoral Training Center in Condensed Matter Physics. CM-DTC-Summer School 2010, Foundations of Condensed Matter Physics, University of St Andrews, Fife, Scotland, August 30-September 8, 2010. CM-DTC-Summer School 2010, Foundations of Condensed Matter Physics, University of St Andrews, Fife, Scotland, August 30-September 8, 2010. , Sep-2010
Papers from Meeting Proceedings	Chaikin PM. "Topological Defects on Hexagonal Lattices in Flat and Curved Space." Statistical Mechanics and Quantum Transport: From Optics to Biology, Haifa, Israel, May 23–27, 2010. Statistical Mechanics and Quantum Transport: From Optics to Biology, Haifa, Israel, May 23–27, 2010. Technion, Haifa, Israel Institute of Technology. , May-2010
Papers from Meeting Proceedings	Chaikin PM. "Experiments With Colloids and Candies: Experimental Geometry." Statistical Mechanics and Quantum Transport: From Optics to Biology, Haifa, Israel, May 23–27, 2010. Statistical Mechanics and Quantum Transport: From Optics to Biology, Haifa, Israel, May 23–27, 2010. Technion, Haifa, Israel Institute of Technology. , May-2010
Papers from Meeting Proceedings	Chaikin PM. "Experiments on Photonic Quasi-Crystals." Statistical Mechanics and Quantum Transport: From Optics to Biology, Haifa, Israel, May 23–27, 2010. Statistical Mechanics and Quantum Transport: From Optics to Biology, Haifa, Israel, May 23–27, 2010. Technion, Haifa, Israel Institute of Technology. , May-2010
Papers from Meeting Proceedings	Chaikin PM. "Some experiments on packing and stacking of nonspherical objects." Optimal configurations on the sphere and other manifolds, Vanderbilt University, Nashville, TN, May 17–20, 2010. Optimal configurations on the sphere and other manifolds, Vanderbilt University, Nashville, TN, May 17–20, 2010. , May-2010